

Gluteal Region

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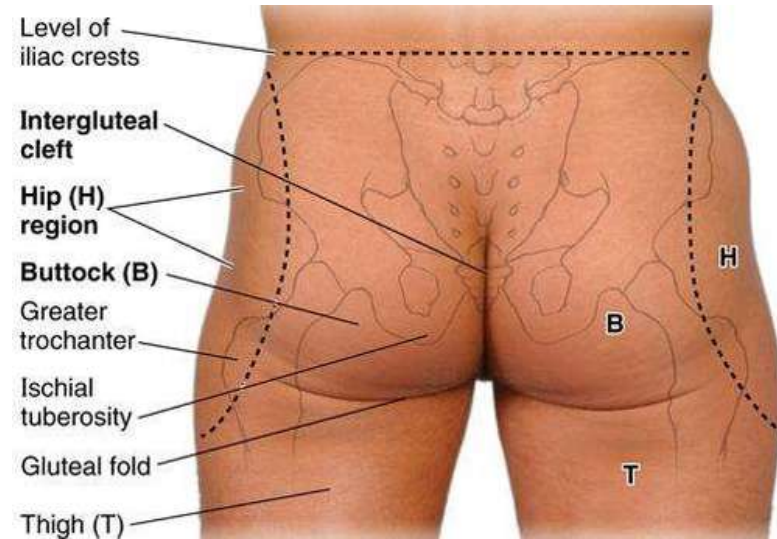
Associate Professor of Anatomy
2025



Where is Gluteal Region?

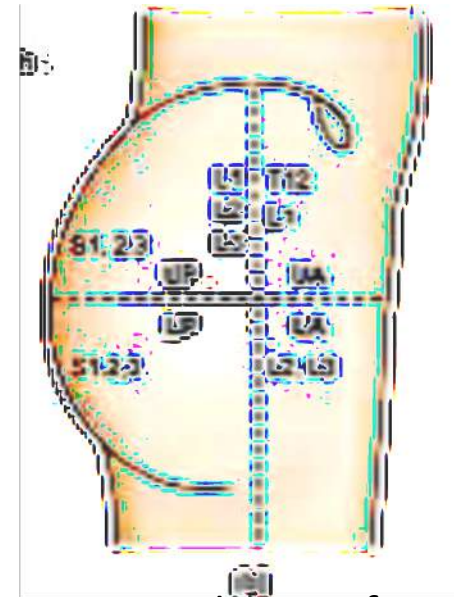
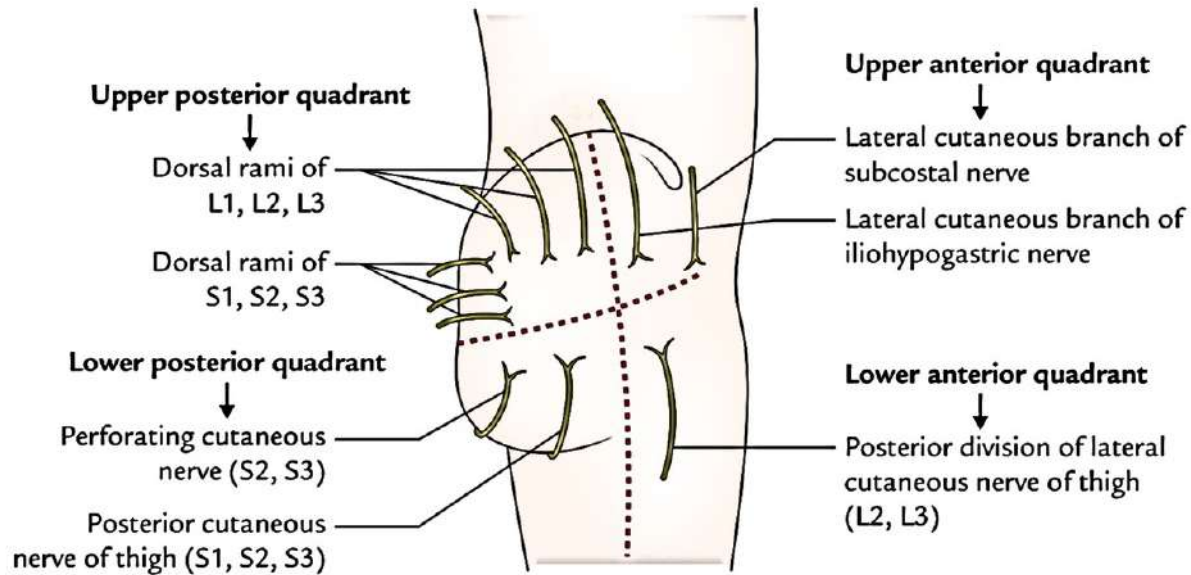


**It is the region
of the buttocks**



Cutaneous nerve	Origin	Area supplied
1- Posterior Rami of L1,2,3 and posterior Rami of S1,2,3	posterior rami	Skin of Upper medial (posterior) quadrant.
2. Lateral branches of the anterior rami of Iliohypogastric (L1) and 12th thoracic nerve (T12) (subcostal nerve)	Anterior rami of L1 (L.Plexus) & T12.	Skin of the Upper lateral (anterior) quadrant.
3- branches from posterior cutaneous nerve of the thigh	Sacral plexus (anterior rami S1,2,3)	Skin of Lower medial(posterior) quadrant.
4- Branches from lateral femoral) cutaneous nerve of the thigh	(Anterior rami L2,3)(L. plexus)	Skin of the lower lateral(anterior) quadrant.

NB: Both anterior & posterior Rami of spinal nerves supply cutaneous nerves to gluteal region.

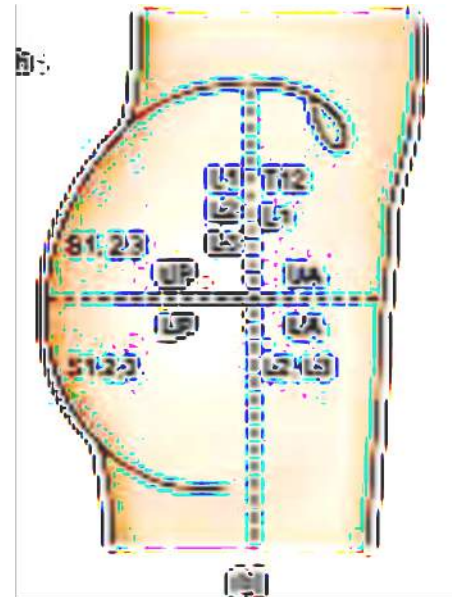
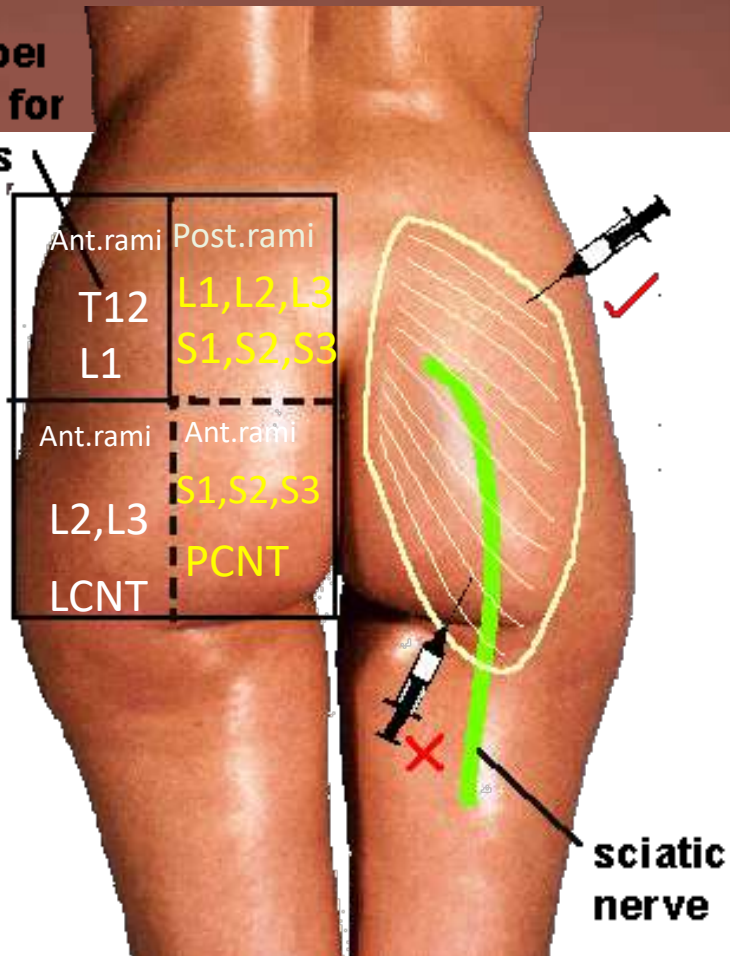


Lateral view of
gluteal region

**outer upper
quadrant for
injections**

Ant.rami	Post.rami
T12	L1,L2,L3
L1	S1,S2,S3
Ant.rami	Ant.rami
L2,L3	S1,S2,S3
LCNT	PCNT

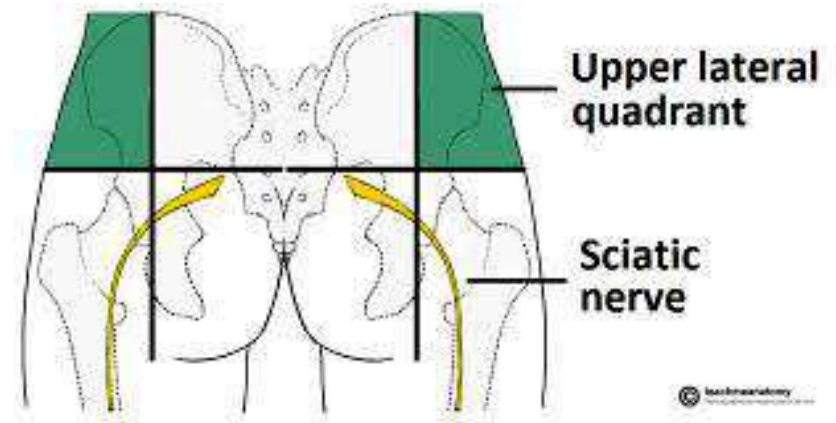
Ant=Lateral
Post=Medial



It is bounded :

Superiorly: by Iliac crest.

Inferiorly : by the inferior border of the gluteus maximus (gluteal fold).



Importance?

- ❑ Contains important ligaments.
- ❑ Contains important muscles that move hip joint.
- ❑ Contains important nerves & vessels.
- ❑ It is the gateway through which nerves pass to supply lower limbs.



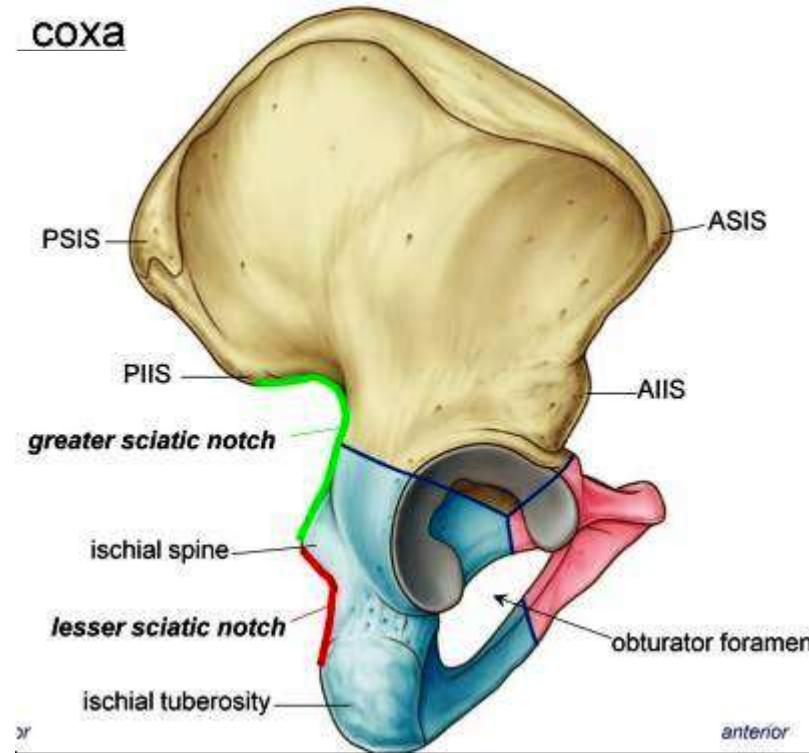
Ligaments of the gluteal region:

Two important ligaments:

❖ **Sacrotuberus ligament.**

❖ **Sacrospinous ligament.**

- **What is the difference between notch & foramen??**



A

Anterior superior iliac spine

Margin of pelvic inlet

Lesser sciatic foramen

Pubic tubercle

Obturator foramen

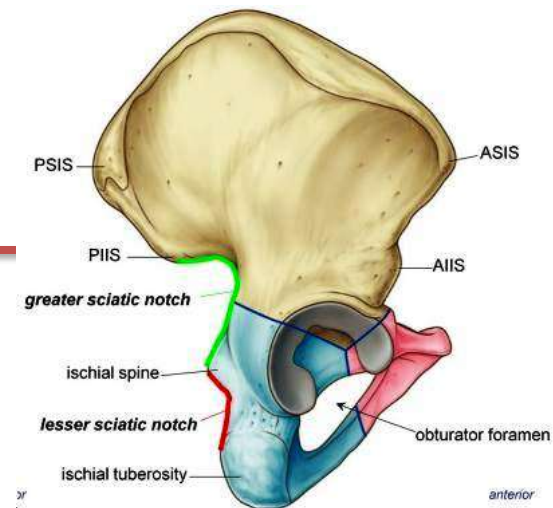
Ischiopubic ramus

Greater sciatic foramen

Sacrospinous ligament

Sacrotuberous ligament

Ischial tuberosity



foramen

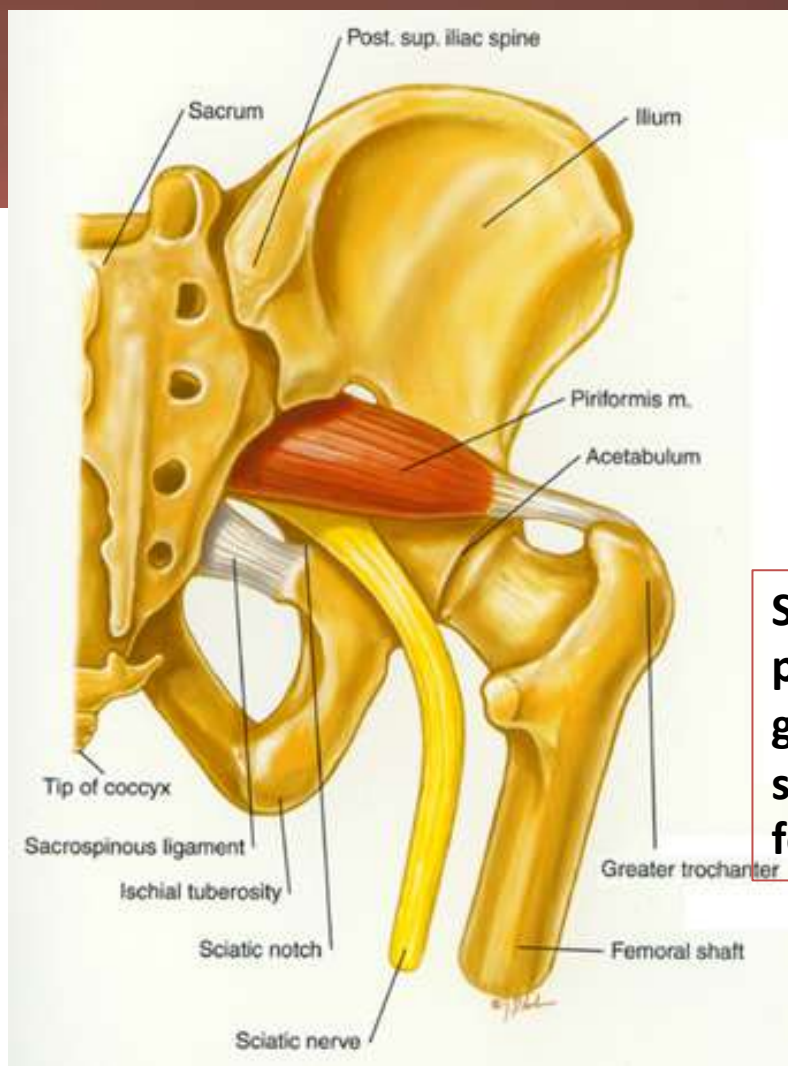
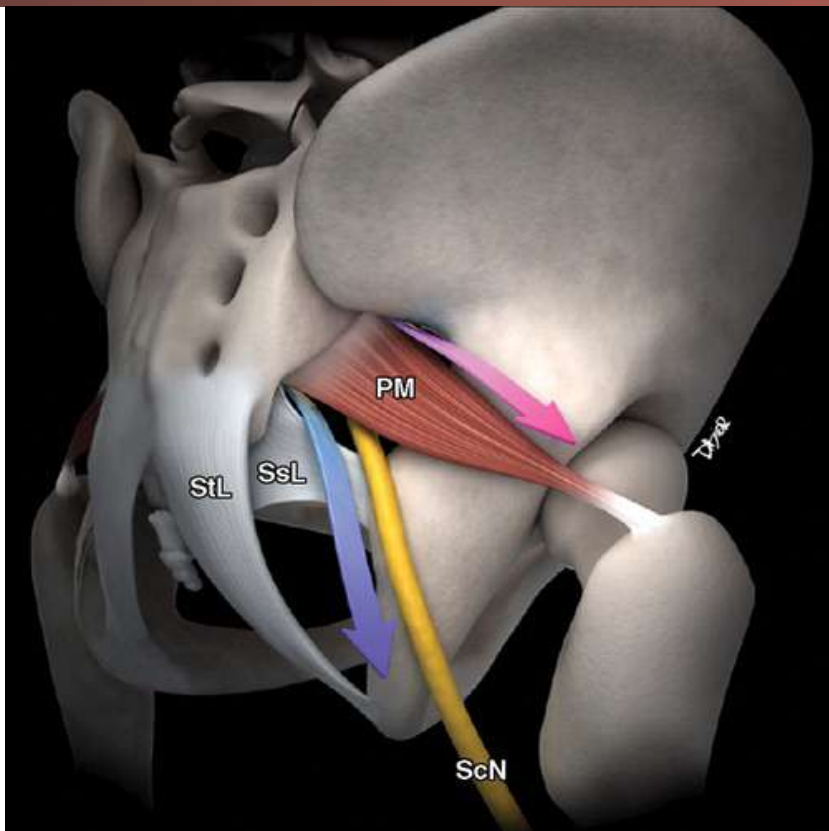
Notch

❖ **Greater sciatic foramen**

- ❖ Formed by greater sciatic notch & Sacrotuberus & sacrospinous ligaments.
- ❖ Provides *Exit from pelvis to gluteal region.*

❖ Structures exit (pass) the GSF :

- ❖ 1-Piriformis muscle
- 2-Sciatic nerve
- 3-Posterior cutaneous nerve of the thigh
- 4-Superior & inferior Gluteal nerves.
- 5-Nerves to obturator internus & nerve to quadratus femoris
- 6-Pudendal nerve.
- 7- Superior & inferior gluteal vessels.
- 8- Internal pudendal vessels.



**Structures
pass through
greater
sciatic
foramen**



Lesser sciatic foramen

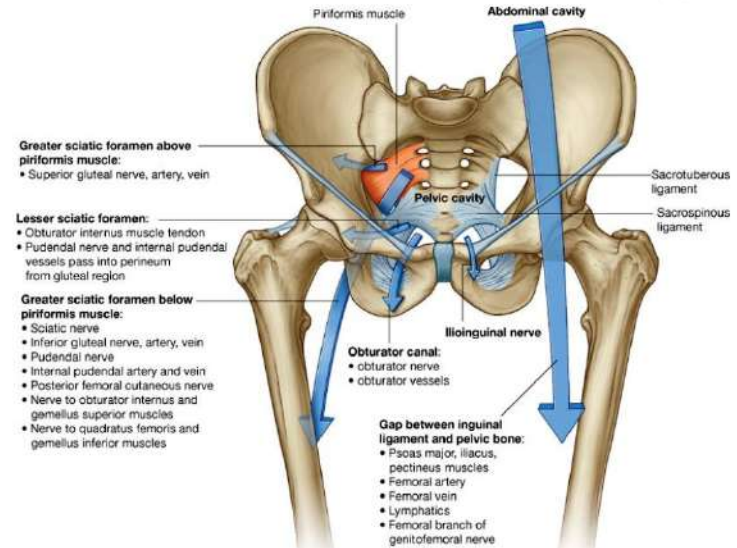
- ❖ Formed by **lesser sciatic notch of hip bone & sacrotuberous & sacrospinous ligaments.**
- ❖ It provides an entrance **into the perineum from the gluteal region.**
- ❖ **Structures pass through it:** مهم
 - 1-Tendon of obturator internus muscle.
 - 2- Nerve to obturator internus.
 - 3- Pudendal nerve.
 - 4- Internal Pudendal vessels.

	<i>Greater sciatic foramen</i>	<i>Lesser sciatic foramen</i>
Muscles	Piriformis	Tendon of Obturator internus
Nerves	➤ Sciatic nerve ➤ Posterior cutaneous nerve of the thigh. ➤ Sup.& inf. Gluteal nerves. ➤ Nerves to obturator internus ➤ nerve to quadratus femoris. ➤ Pudendal nerve.	➤ Nerve to Obturator internus. ➤ Pudendal nerve.
Vessels	➤ Internal Pudendal vessels	➤ Internal Pudendal vessels

Gateway to the lower Limb:

❑ Four major routes by which structures pass from the abdomen and pelvis into and out of the lower limb:

- 1- Obturator canal
- 2- GS foramen
- 3- LS foramen
- 4- Gap between inguinal ligament and antero-superior margin of pelvis.



Greater sciatic foramen above piriformis muscle:

- Superior gluteal nerve, artery, vein

Lesser sciatic foramen:

- Obturator internus muscle tendon
- Pudendal nerve and internal pudendal vessels pass into perineum from gluteal region

Greater sciatic foramen below piriformis muscle:

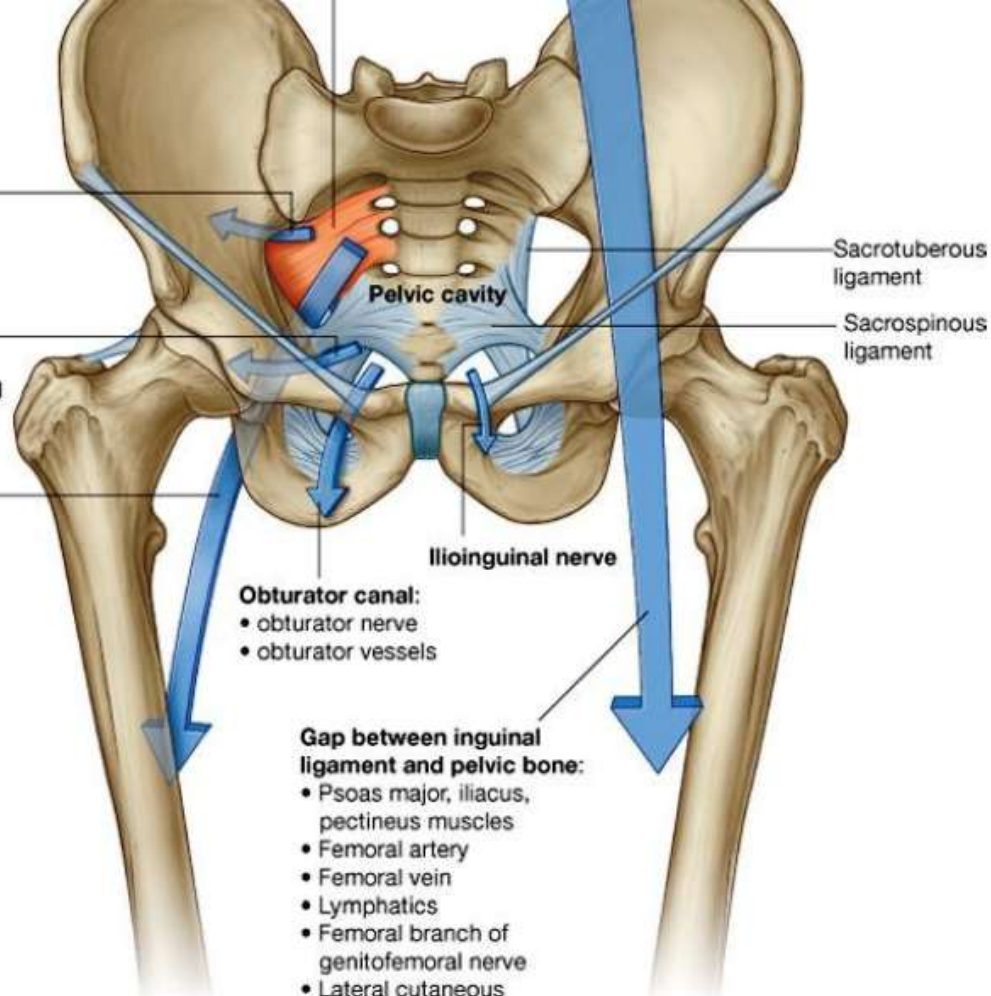
- Sciatic nerve
- Inferior gluteal nerve, artery, vein
- Pudendal nerve
- Internal pudendal artery and vein
- Posterior femoral cutaneous nerve
- Nerve to obturator internus and gemellus superior muscles
- Nerve to quadratus femoris and gemellus inferior muscles

Obturator canal:

- obturator nerve
- obturator vessels

Gap between inguinal ligament and pelvic bone:

- Psoas major, iliacus, pectineus muscles
- Femoral artery
- Femoral vein
- Lymphatics
- Femoral branch of genitofemoral nerve
- Lateral cutaneous nerve of thigh
- Femoral nerve

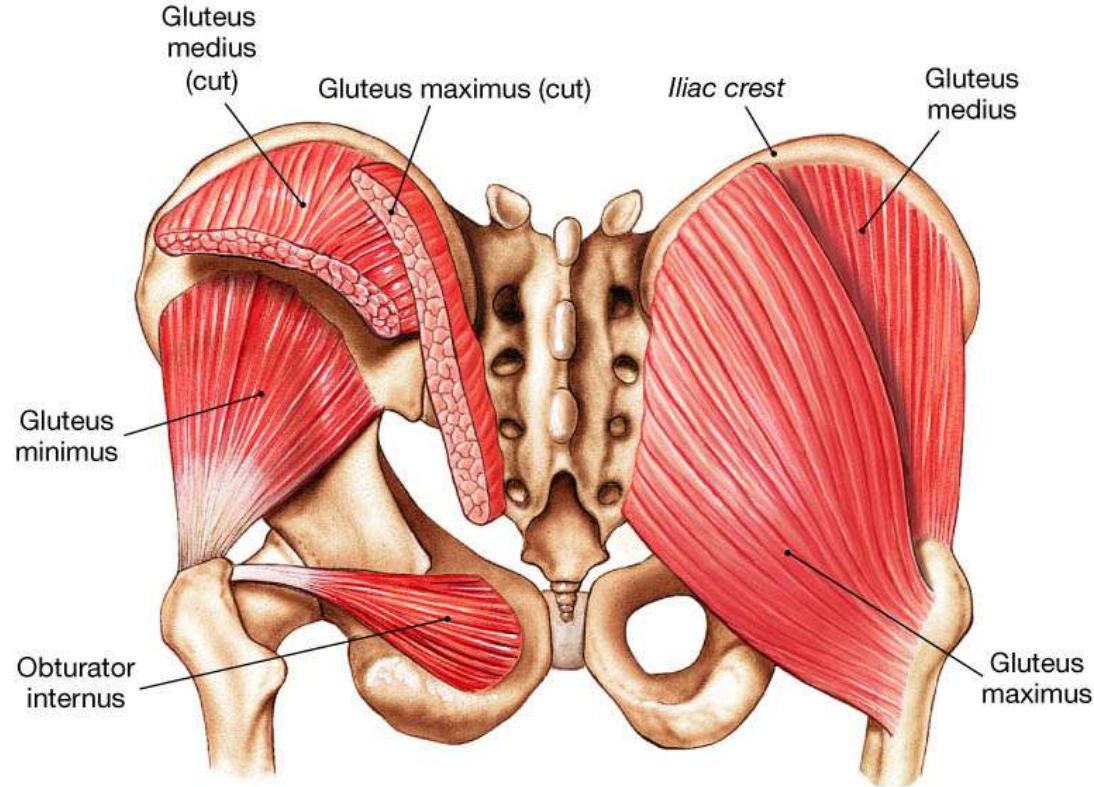


Muscles of the Gluteal region

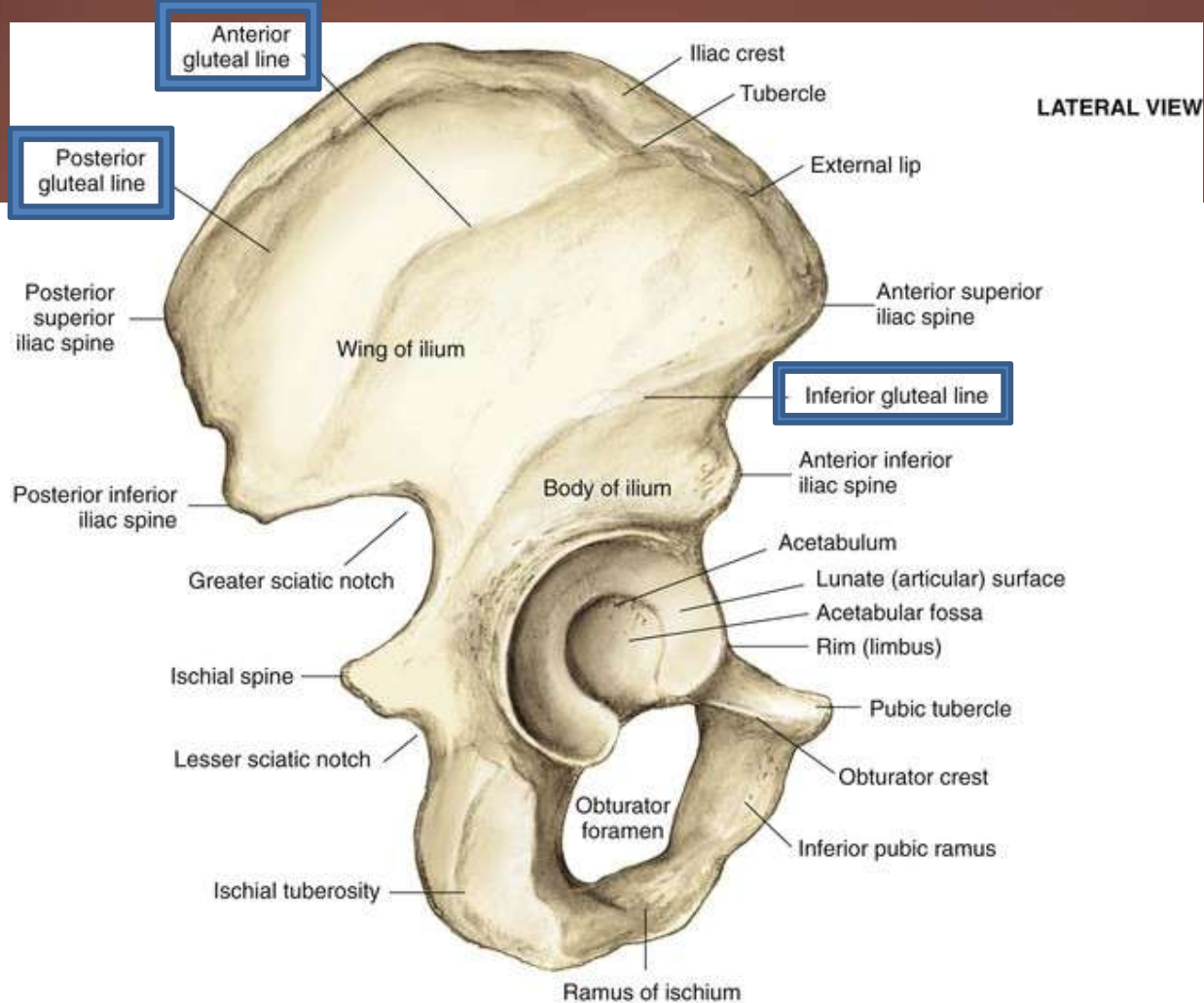
A- Superficial large group (4)

- **Gluteal** muscles
- *Extensors* and *Abductors* of hip joint.

B-Deep small group (5) **Lateral rotators** of femur.



(c) Gluteal and lateral rotators, posterior view



❑ Superficial group:

1-Gluteus Maximus:

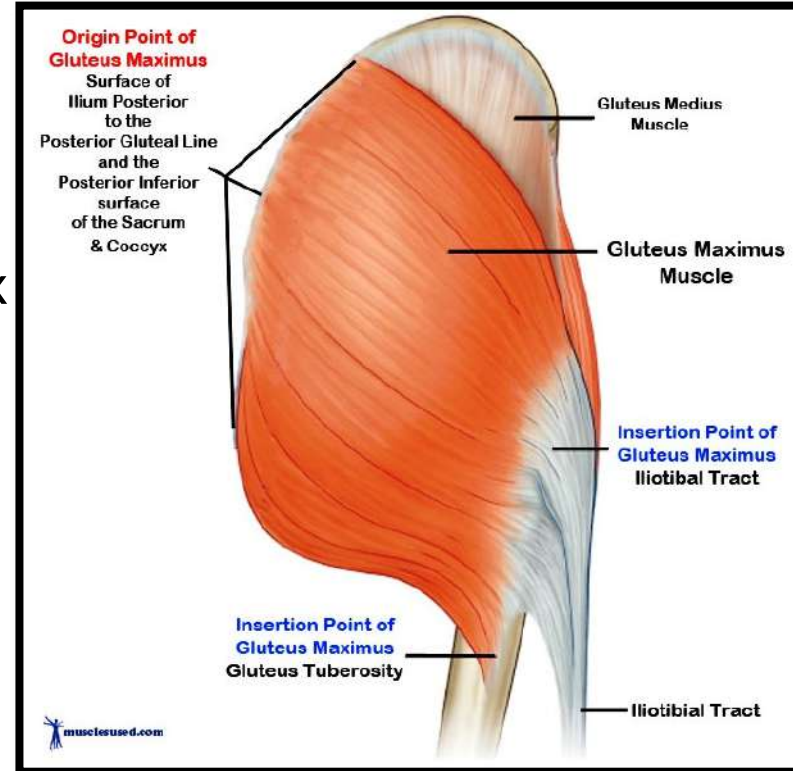
❖ Largest muscle in the body, lies superficial in gluteal region, responsible for prominence of buttock.

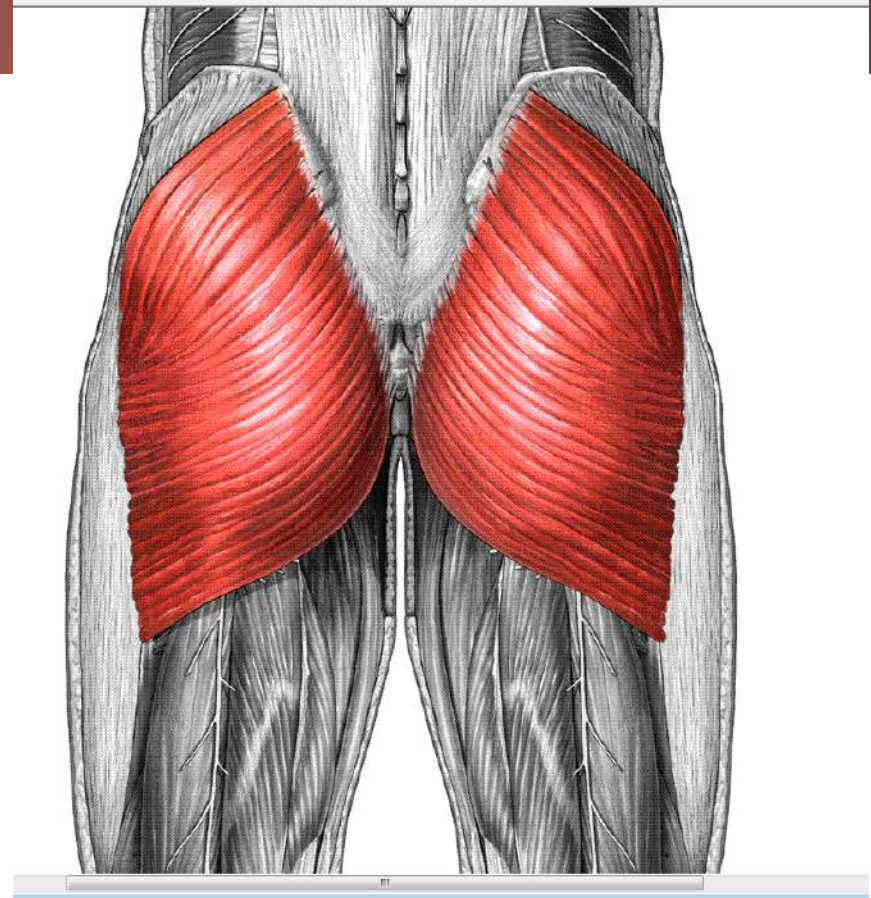
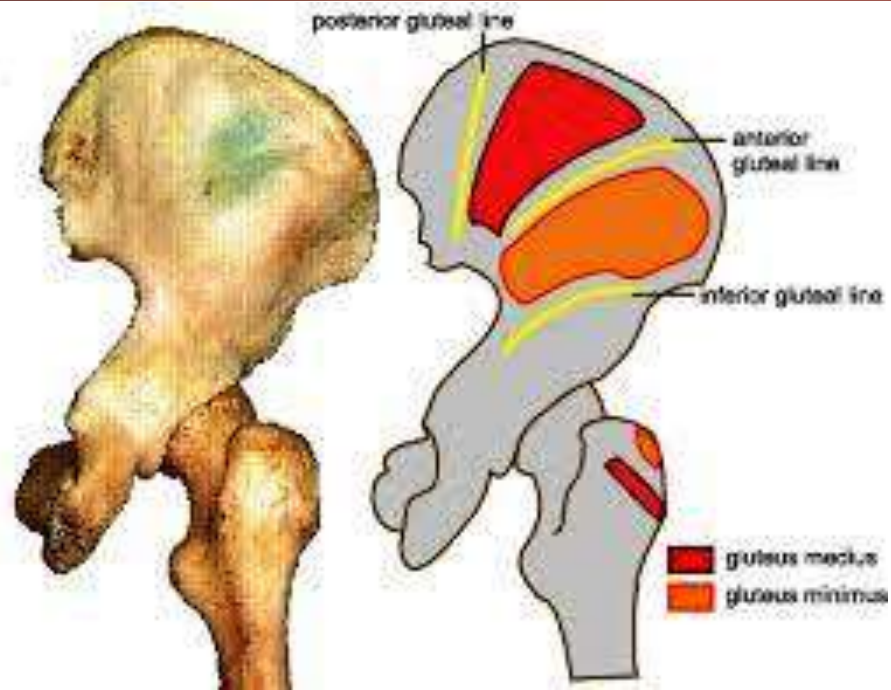
O : 1)Ext. surface of ilium, **behind** post.gluteal line 2) back of sacrum & coccyx 3) Sacrotuberous ligament.

I : **Most** fibers inserted into **iliotibial tract**, some deeper fibers inserted into **gluteal tuberosity of femur**.

N : Inferior gluteal nerve.

A : It is a **powerful extensor** of hip joint+ assists in **lateral rotation** of hip joint+ it acts with T. F latae to **maintain knee joint** in extension through **iliotibial tract** in erect posture.

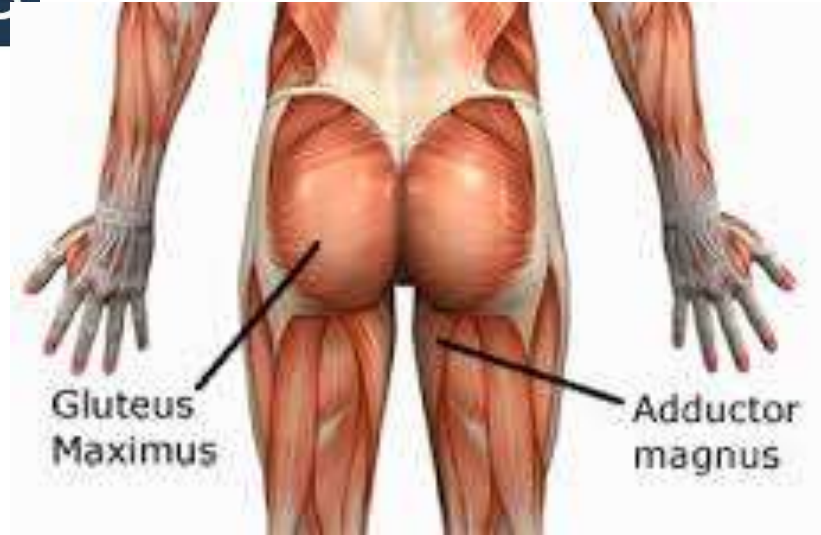




Gluteus Maximus



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2-Gluteus Medius

Thick, **fan-shaped** muscle, & covered by G.maximus.

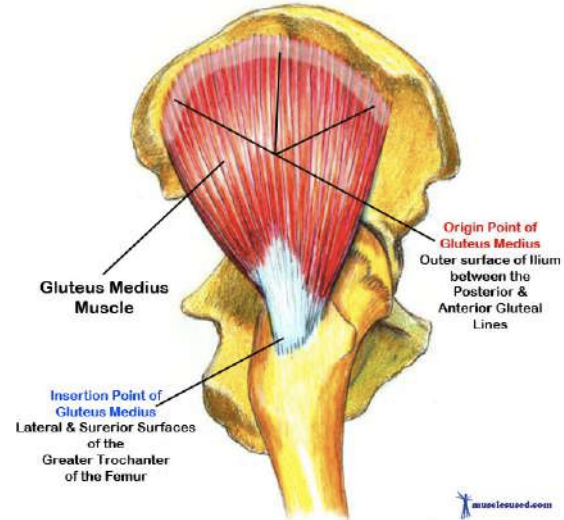
O : Ext.surface of ilium **between anterior & posterior gluteal lines.**

I : into **lateral surface of greater trochanter.**

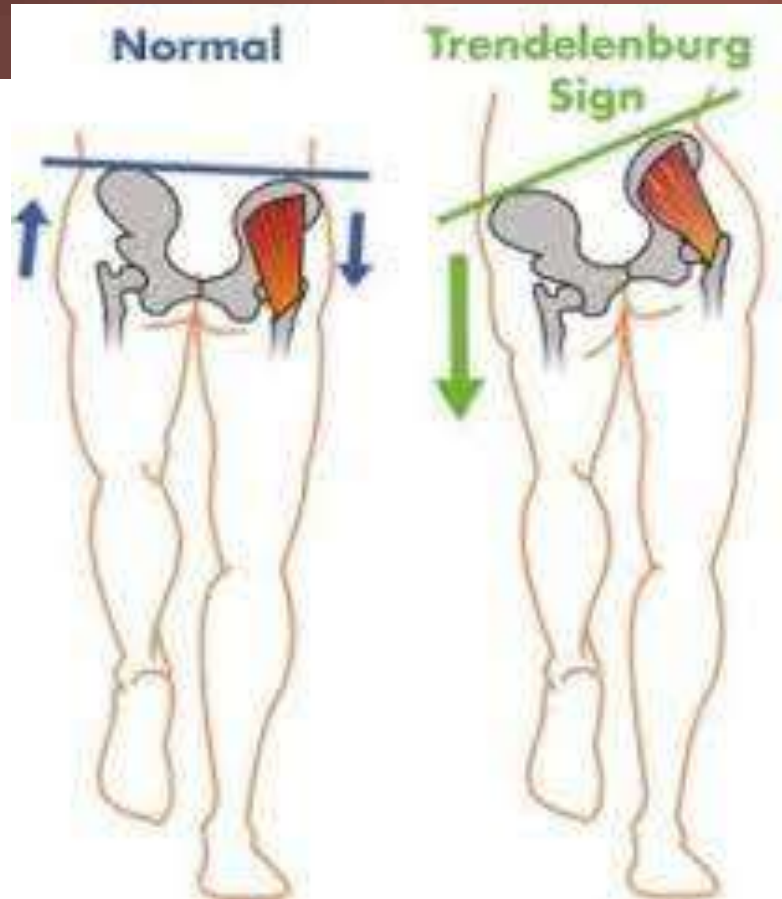
N : **Superior gluteal nerve.**

A : 1)Acting with G.minimus, it powerfully **Abduct the thigh** at hip joint. 2)**Its ant.fibers medially rotates the thigh.**

3)It also **steadies the pelvis** so that it does not sag when the foot on the opposite side is raised(e.g during walking).



Applied



Superior gluteal nerve injury:
causes hip abductor weakness,
leading to a characteristic
waddling gait and a **positive
trendelenburg sign**, where the
pelvis dips on the unsupported
side when standing on the
injured leg.

Waddling gait video

<https://youtu.be/R4dl4nrsVd8?si=3JXNEe23BGEVvglo>

3-Gluteus Minimus

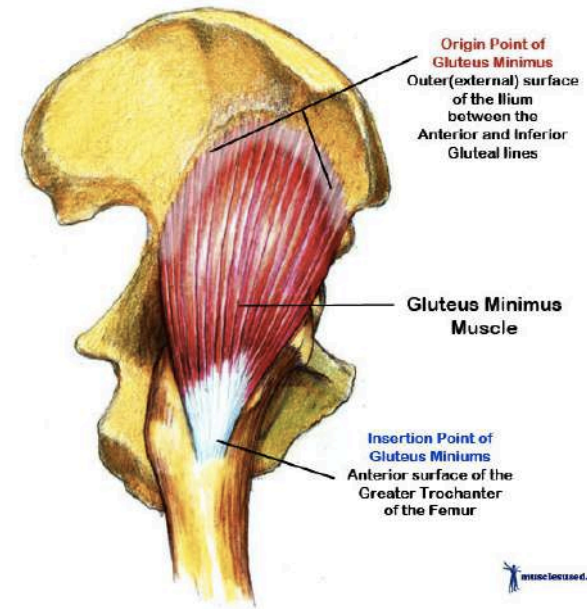
Fan shaped & lies deep to G. medius.

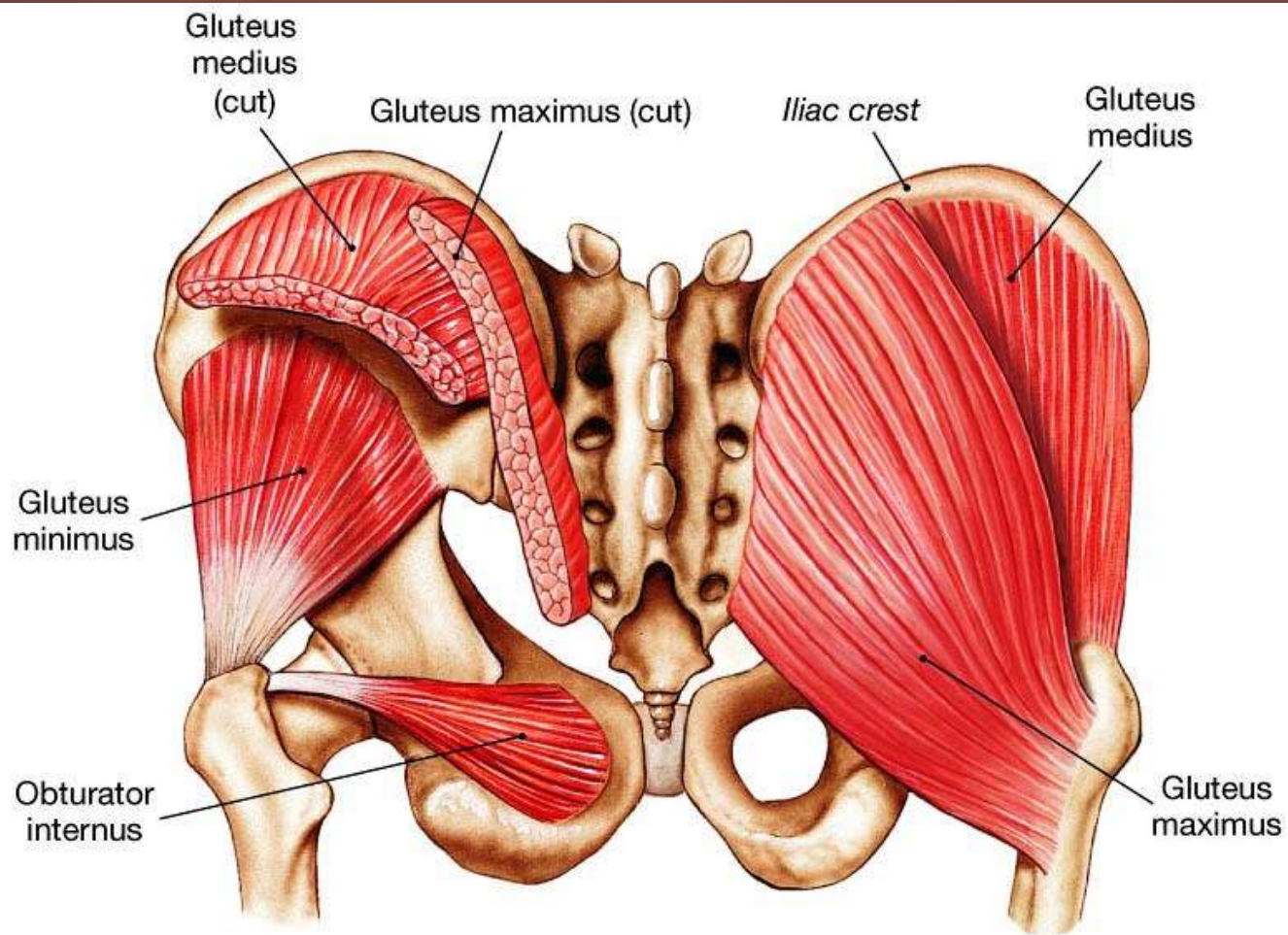
O : Ext. surface of ilium between anterior & inferior gluteal lines.

I : into **anterior surface of greater trochanter**.

N : **Superior gluteal nerve**.

A : acting with G. medius, powerfully **abducts** the thigh at hip joint.+ its **anterior fibers medially rotate the thigh**.





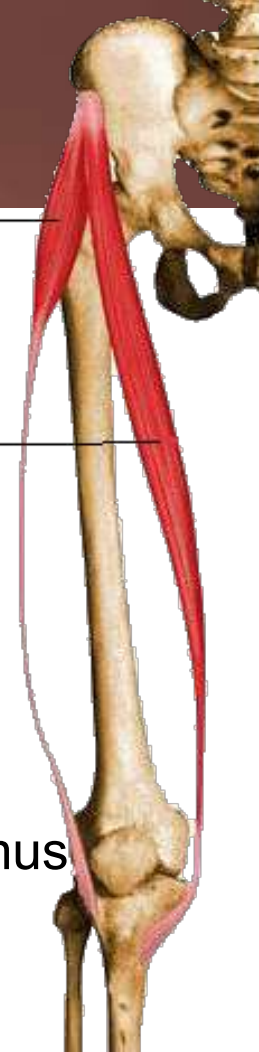
(c) Gluteal and lateral rotators, posterior view

4-Tensor fasciae latae

- ❖ fusiform M. lies
 - ❖ on the lateral side of front of the thigh, enclosed in fascia lata
- O** : from **ASIS** & iliac crest.
- I** : into Iliotibial tract.
- N** : *Superior gluteal nerve.*
- A** : It exerts traction on iliotibial tract & so it assists gluteus maximus in extending the knee joint in standing position.

Tensor fascia lata

Sartorius



Always remember: مهم

❖ Superior gluteal nerve supplies 3 muscles

1- gluteus medius

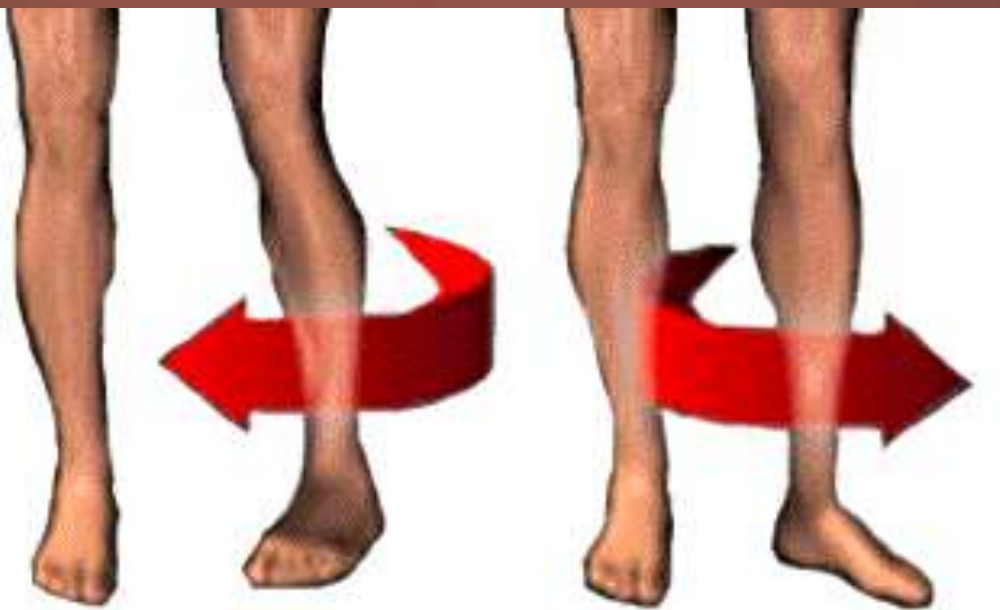
2-gluteus minimus

3-Tensor fasciae latae

❖ Inferior gluteal nerve supplies 1 muscle :

Gluteus maximus





Medial Rotation

Lateral Rotation

Deep short group of muscles(5)

1-Piriformis

2-Gemellus Superior

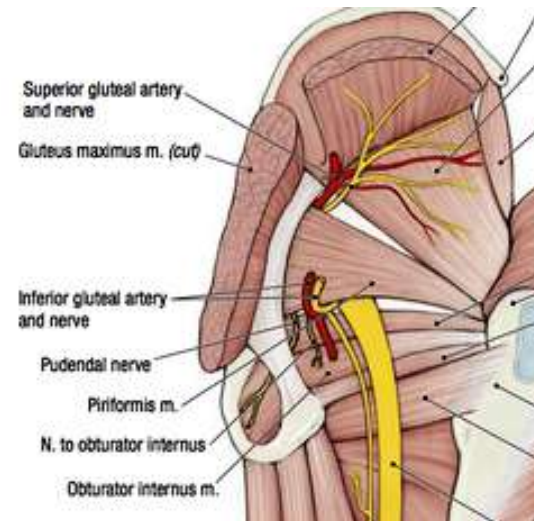
3-Gemellus inferior

4-Obturator internus

5-Quadratus Femoris

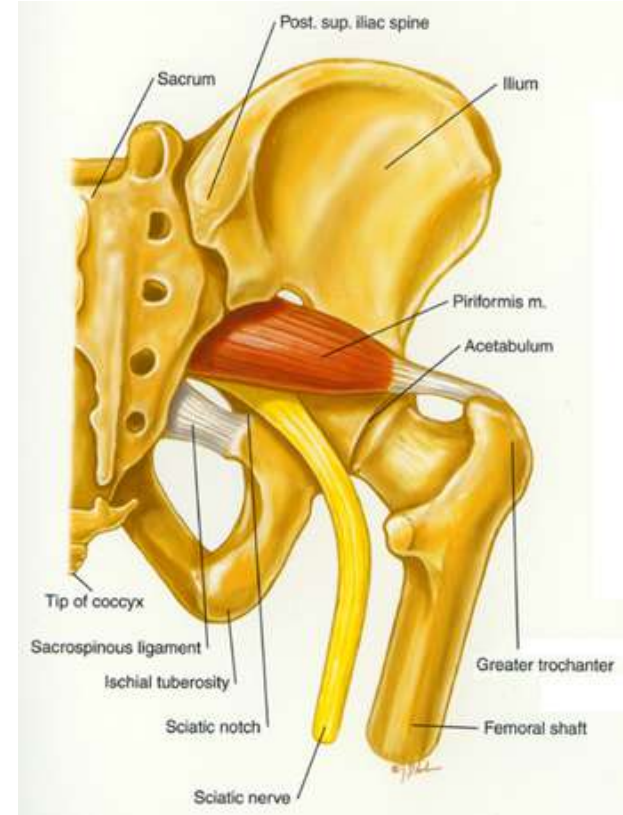
Lateral
Rotator of
the thigh

N.B. **Obturator externus** muscle studied with medial thigh, is also **lateral rotator** of the thigh, its insertion is visible only during dissection of gluteal region. Most effective as a **lat.rotator** when hip joint is flexed.



1-Piriformis muscle

- Is the **most superior** of the deep group.
- Is the muscle of the pelvic wall and of gluteal region.
- Conical-shaped.
- Important **landmark**, because it divides the greater sciatic foramen into two regions, one above & one below the piriformis.

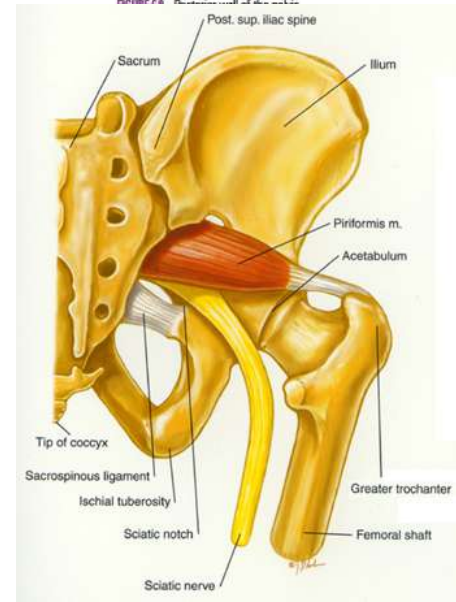
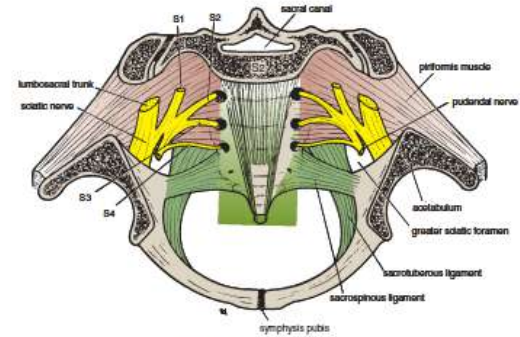


O: from the **anterior surface** of the 2nd, 3rd, and 4th Sacral vertebrae within the pelvis.

I : the fibers passes through the **greater sciatic foramen** and are attached to the **upper border** of the greater trochanter of femur.

N : **Anterior rami of 1st & 2nd Sacral nerves** in pelvis.

A : **Lateral rotator** of the thigh at the hip joint.



2-Obturator internus

❖ Is a muscle of **pelvis wall** and **gluteal region**.

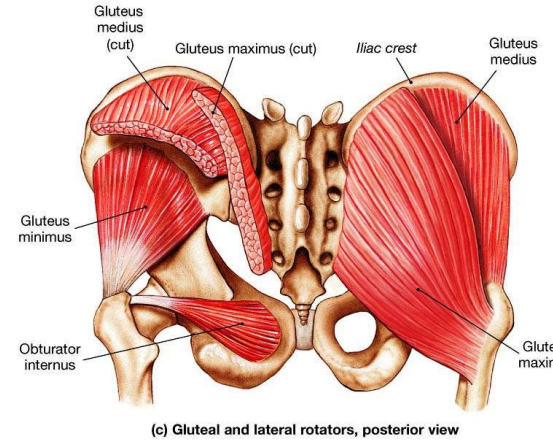
❖ Flat fan shaped muscle.

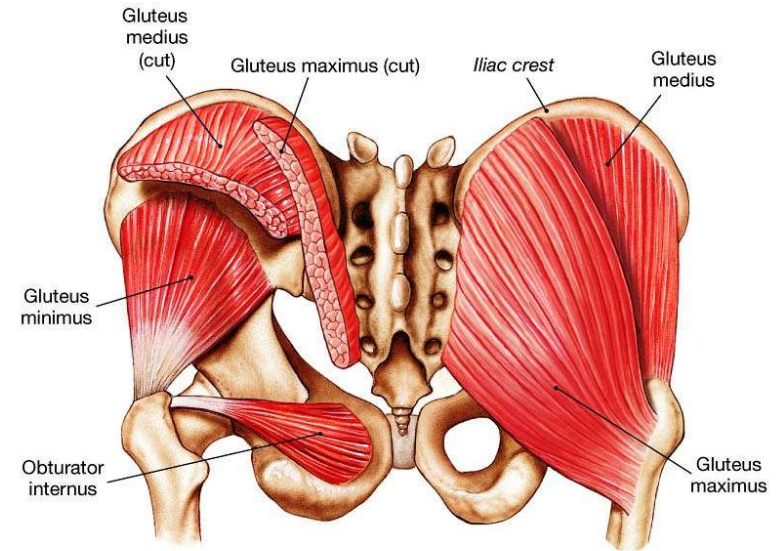
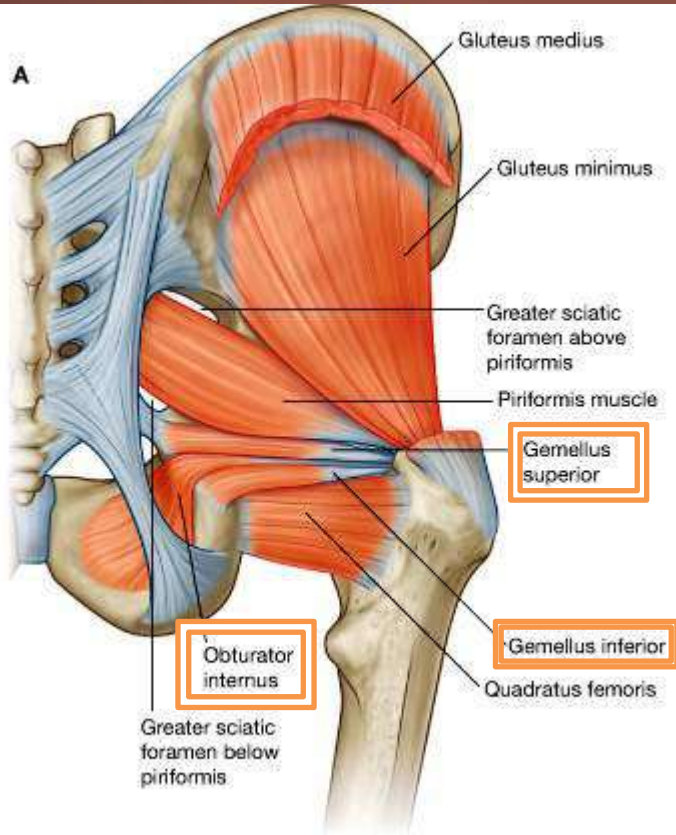
O : from the **pelvic surface of obturator membrane** & surrounding bones.

I : The tendon passes out of pelvis through **lesser sciatic foramen**, joined by Sup. gemellus & Inf. gemellus and common tendon is inserted into **upper border** of the greater trochanter.

N : Nerve to **obturator internus** from sacral plexus.

A : **Lateral rotator** of the thigh at the hip joint.

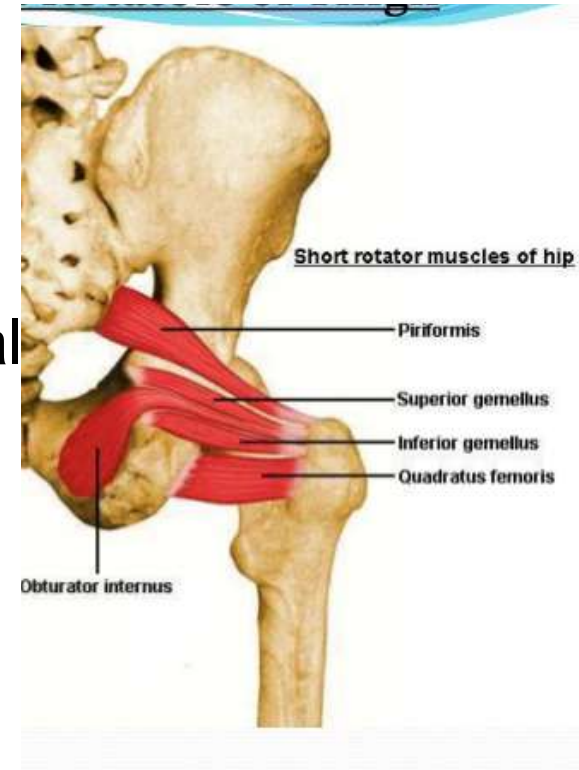




(c) Gluteal and lateral rotators, posterior view

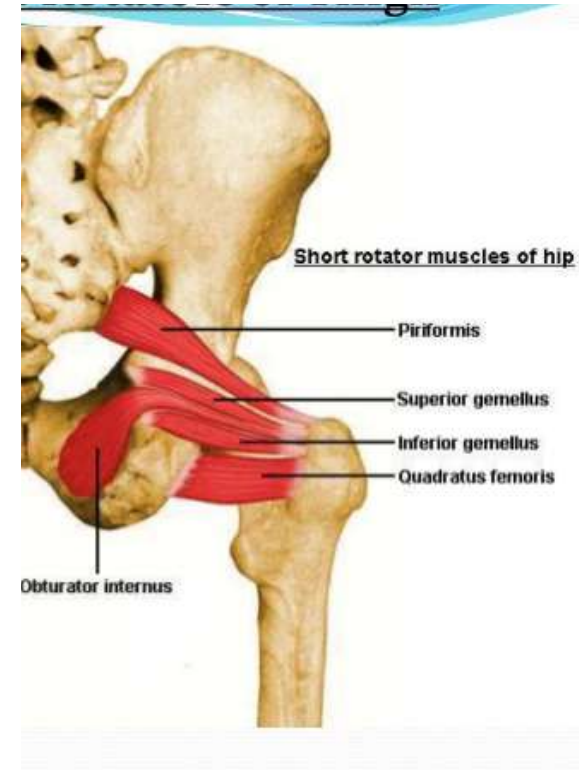
3-Gemellus Superior

- O:** Small muscle from **ischial spine**.
- I:** With **common tendon** with obturator internus.
- N:** nerve to **obturator internus** from Sacral plexus
- A:** **Lateral rotator** of the thigh at the hip joint.



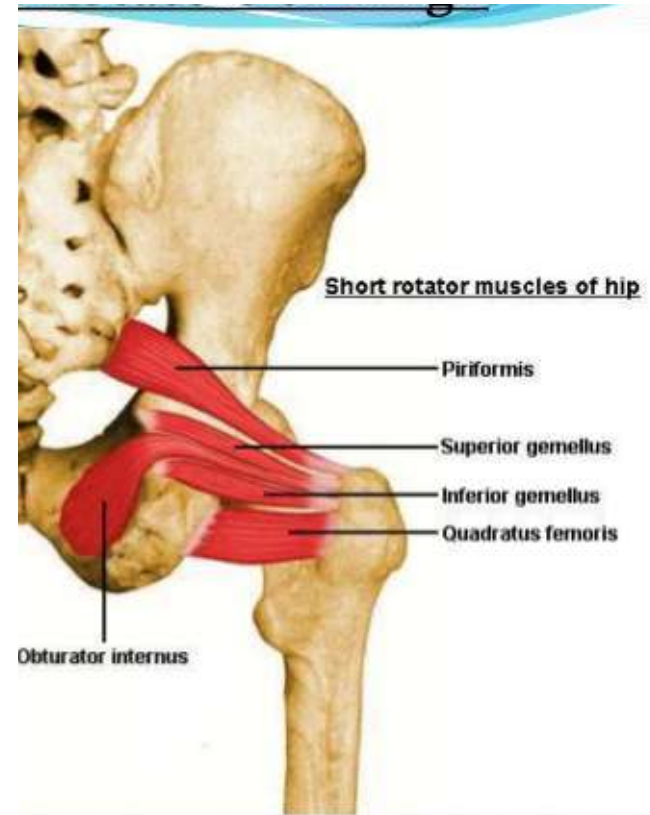
4-Gemellus Inferior

- O:** from **upper border of ischeal tuberosity**.
- I:** With **common tendon** with obturator internus.
- N:** nerve to **quadratus femoris** from sacral plexus.
- A:** **Lateral rotator** of the thigh at the hip joint.



5- Quadratus femoris (quadrilateral –shaped muscle).

- O** : from **lateral border of ischial tuberosity**
- I** : into **quadrate tubercle** on **intertrochanteric crest of the femur.**
- N** : nerve to **quadratus femoris** from **sacral plexus.**
- A** : **Lateral rotator** of the thigh at the hip joint.



Short Lateral Rotators of Thigh

Gemellus superior

Origin-spine of ischium

Insertion-tendon of OBT int

Nerve- to OBT internus

Gemellus inferior

Origin-ischial tuberosity

Insertion-tendon of OBT internus

Nerve-to Quadratus femoris

Quadratus femoris

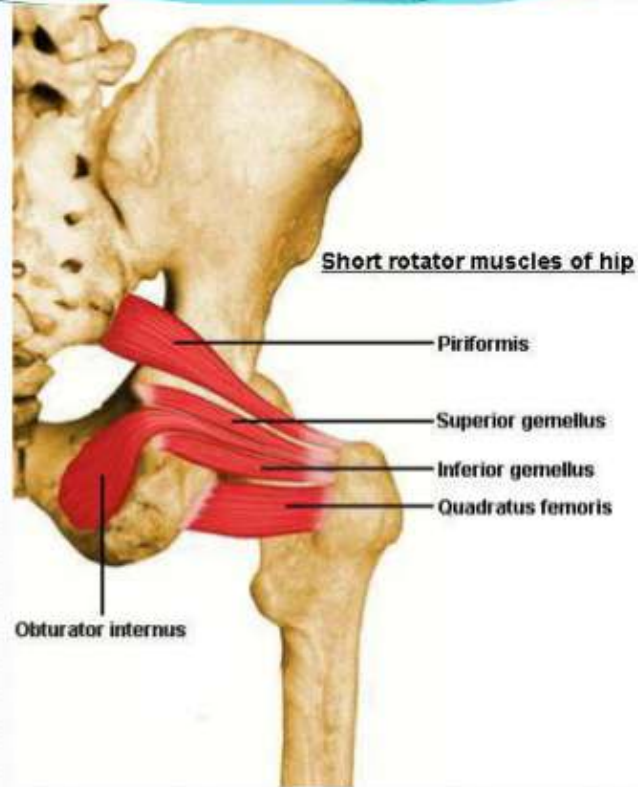
Origin-ischial tuberosity

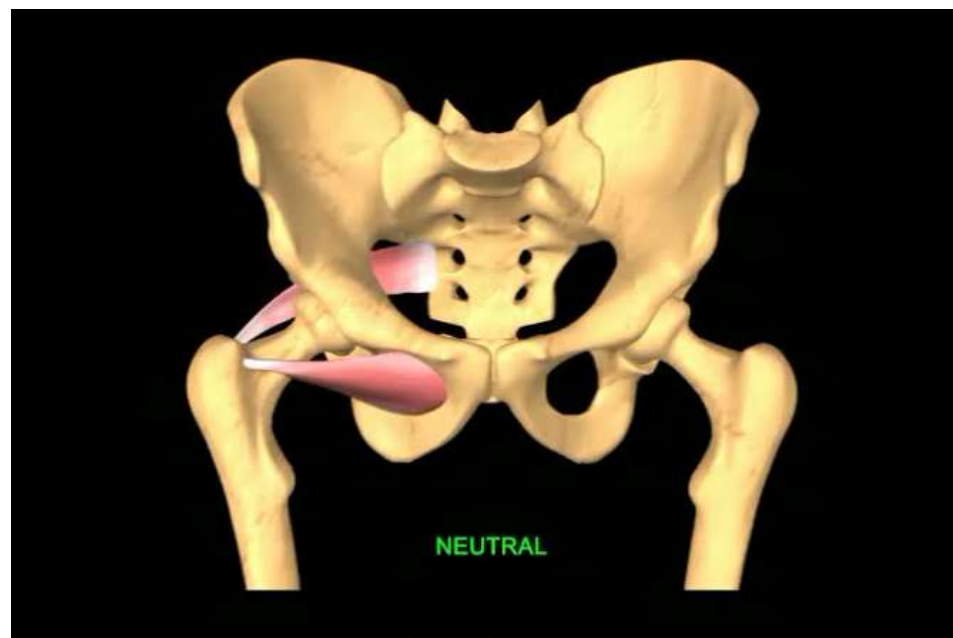
Insertion-quadratus tubercle

Nerve-sacral plexus

Obturator internus

Piriformis





Greater trochanter:

Gluteus Medius: into lateral surface of greater trochanter.

Gluteus minimus: : into ant.surface of greater trochanter

Piriformis: upper border of the greater trochanter.

Obturator internus: inserted into upper border of the greater trochanter.

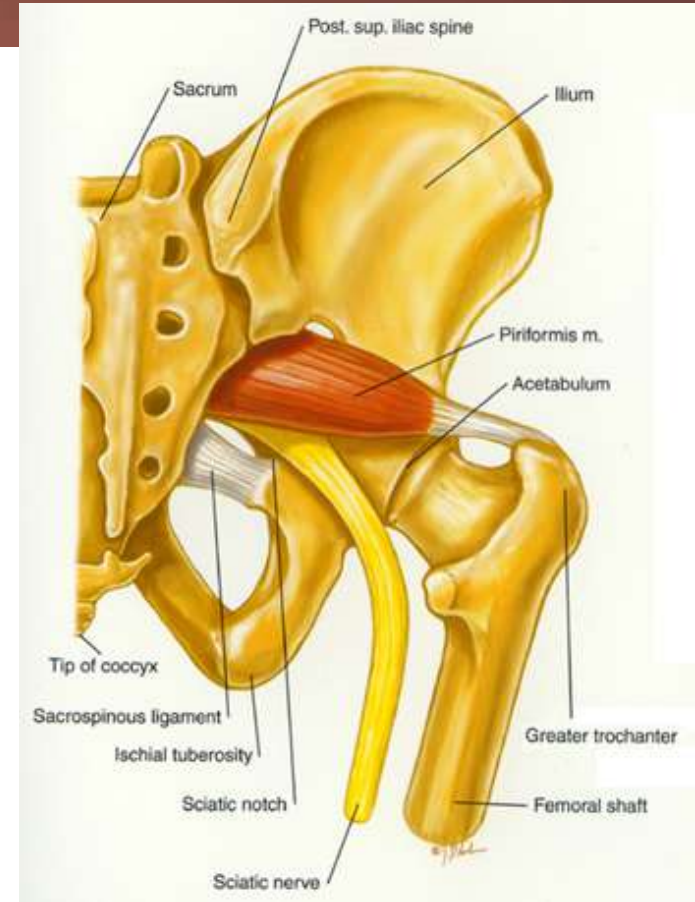
- ❖ All the short lateral rotators are supplied by branches from sacral plexus *except* **Obturator Externus** by obturator nerve (from lumbar plexus).
- ❖ All the short muscles have the same action which is lateral rotation of thigh at hip joint.
- ❖ All the short lateral rotators are inserted into of upper border of **greater trochanter** Except Quadratus femoris inserted into **quadarate tubrecle** on intertrochanteric crest.

- 1) **Flexors of the thigh:** Pectineus, iliacus, psoas, sartorius, rectus femoris, gracilis.
- 2) **Extensors of the thigh:** hamstring portion of adductor magnus. & **gluteus maximus.**
- 3) **Lateral rotators of the thigh:** Obturator externus, sartorius, (adductor longus, brevis & magnus.) , **Piriformis, Gemellus Superior , Gemellus inferior, Obturator internus, Quadratus Femoris.**
- 4) **Medial rotators of the thigh:** iliacus, psoas. + **anterior fibers of gluteus medius.**
- 5) **Adductors of the thigh:** Pectineus, adductor longus, adductor brevis, adductor magnus & gracilis.
- 6) **Abductor of the thigh:** Sartorius, **Gluteus medius and gluteus minimus**
- 7) **Flexors of the knee:** Sartorius , Gracilis
- 8) **Extensors of the knee:** Quadriceps femoris

Nerves of the gluteal region:

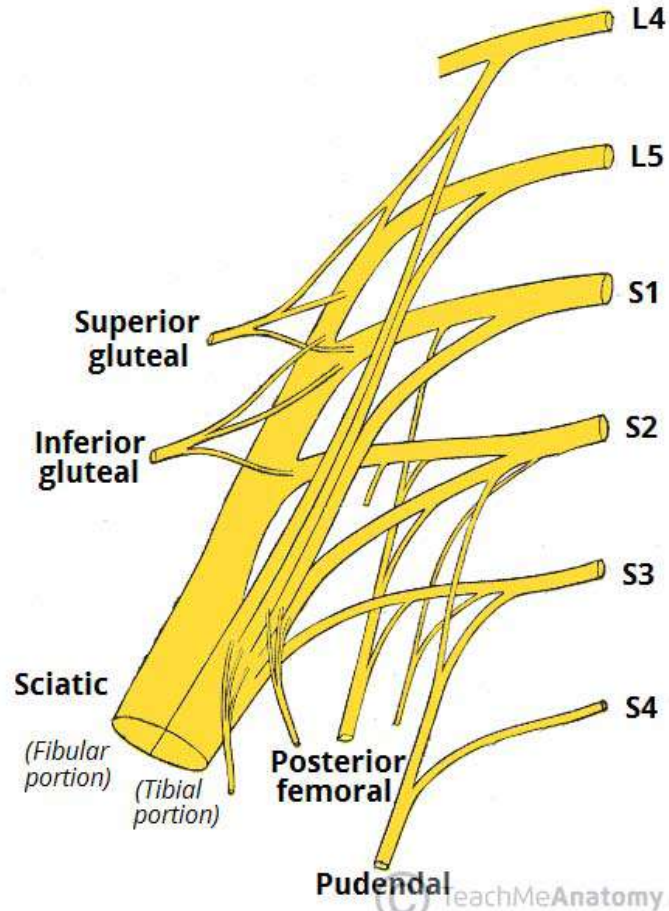
SCIATIC NERVE

عرق النسا.

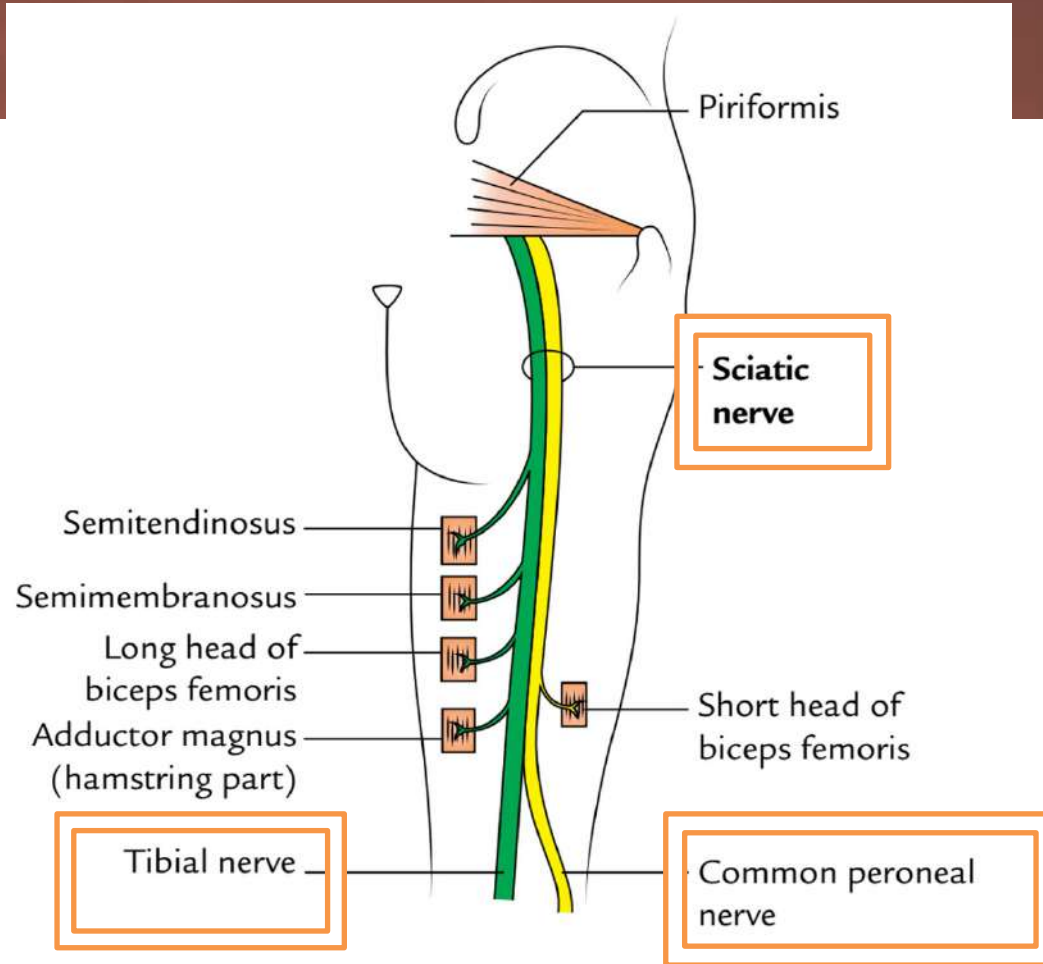


1-Sciatic nerve

- It is the **largest nerve in** the body. Consists of tibial & common peroneal nerves bound together by fascia.
- **Origin:** from **Sacral plexus** (**L**4 ,5 and **S** 1,2 &3) in the pelvis.
- **Course:**
 - It emerges from pelvis through lower part of **greater sciatic foramen**, below the piriformis.
 - It is related **posteriorly to posterior cutaneous nerve of the thigh& the gluteus maximus ??**
 - It leaves gluteal region by passing deep to **long head of biceps femoris** to enter back of the thigh.
- **End:** in the lower third of the thigh, ends by dividing into **tibial &common peroneal nerves**.
- **Branches:** usually it has no branches in gluteal region. 



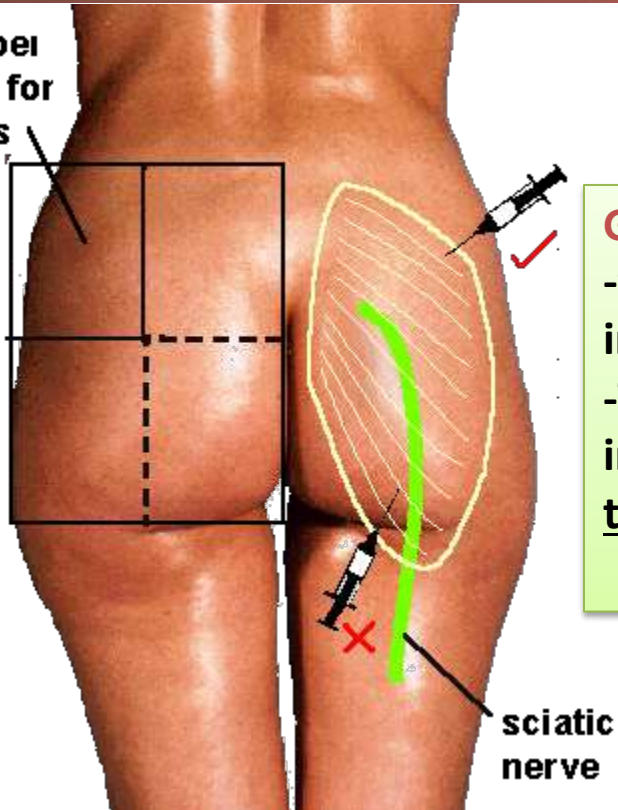
Sacral
plexus



Clinical note:



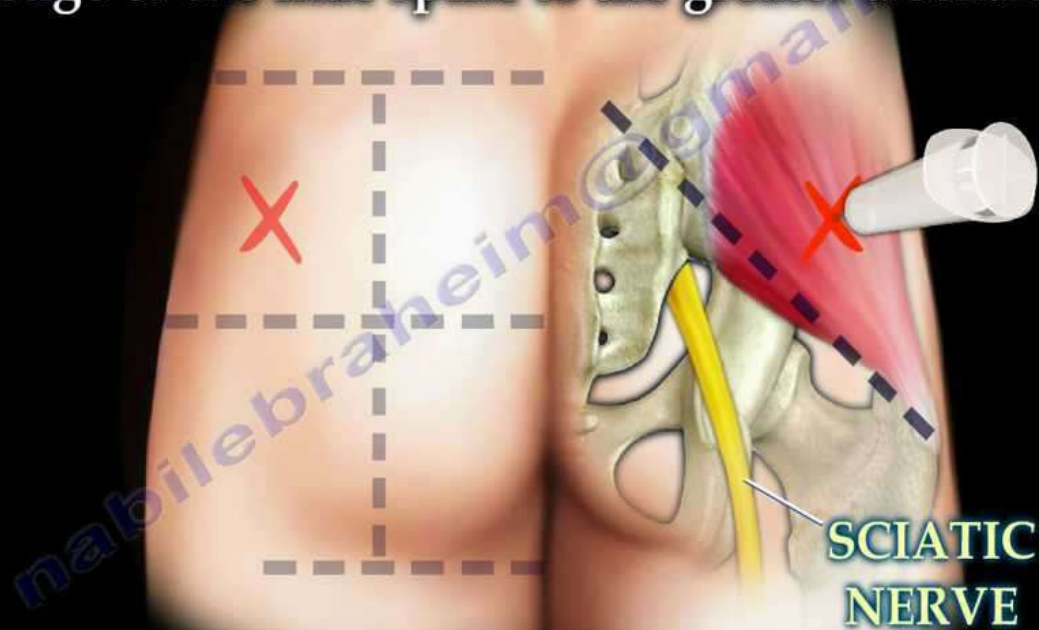
outer upper
quadrant for
injections



Gluteus maximus and intramuscular injection-
-the great thickness of this muscle make it ideal for intramuscular injection.
-To avoid injury to underlying sciatic nerve , the injection should be given on upper outer quadrant of the buttock.

Buttock Injection

A line can be drawn from the posterior superior edge of the iliac spine to the greater trochanter.



2-Posterior cutaneous nerve of the thigh

- Branch of **sacral plexus (S1,2,3)**
- Enters gluteal region through **the lower part of greater sciatic foramen, below piriformis.**
- Runs downward **on post. surface of sciatic nerve** and runs down the back of the thigh under the deep fascia. مهم
- In the popliteal fossa it supplies the skin.

❖ Branches (cutaneous):

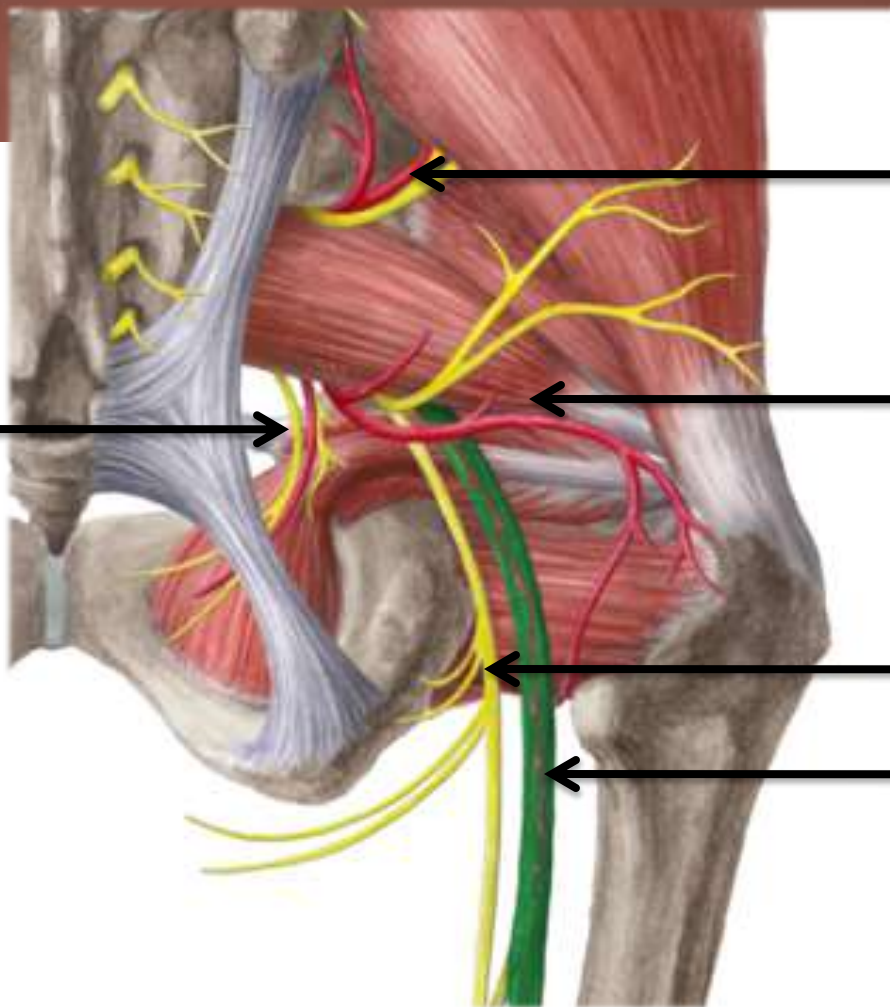
- 1. Gluteal branches:** to the skin of **lower posterior quadrant of the buttock.**
- 2. Perineal branches:** to the skin of the **back of scrotum or labium majus.**
- 3. Cutaneous Branches:** to the **back of the thigh & upper part of leg (popliteal fossa).**



3-Superior gluteal nerve:

- branch of **sacral plexus**
- It enters gluteal region above the piriformis.
- Supplies **G.med, G.min & T.F.latae.**

Internal
pudendal artery and
pudendal
nerve



Superior gluteal
vessels & nerve

Inferior gluteal
vessels & nerve

Posterior cutaneous
nerve of thigh

Sciatic nerve

4-Inferior gluteal nerve:

- Branch of **sacral plexus**.
- It enters gluteal region below the piriformis.
- It supplies only **Glu.Maximus**.

5-Nerve to quadratus femoris:

- Branch of **sacral plexus**.
- It enters gluteal region below the piriformis.
- Supplies ; **Quadratus femoris& inferior gemellus**

6-Pudendal nerve & nerve to obturator Internus.

- Branches of **sacral plexus**.
- They enter gluteal region below the piriformis.
- They immediately reenter the pelvis with **internal pudendal artery** through lesser sciatic foramen.
- Pudendal nerve supplies structures in perineum.
- They Lie over sacro-spinous ligament.

3-Other nerves of sacral plexus

nerve	origin	relation	branches
Superior gluteal nerve	From Sacral plexus (L4,5,S1)	Leaves pelvis through upper part of greater sciatic foramen Above piriformis.	Gluteus medius Gluteus minimus Tensor fascia lata
Inferior gluteal nerve	From Sacral plexus (L5,S1,2)	Leaves pelvis through lower part of greater sciatic foramen below piriformis.	Gluteus maximus
Nerve to quadratus femoris	From sacral plexus (L4,5,S1)	Leaves pelvis through lower part of greater sciatic foramen	Quadratus femoris inferior gemellus
Pudendal nerve(S2,3,4)	From sacral plexus	Leave pelvis through lower part of greater sciatic foramen below piriformis.	Pudendal nerve :supplies structures in perineum.
nerve to obturator internus(L5,S1,2)			Nerve T.O.I. : supplies obturator internus and superior gemellus.

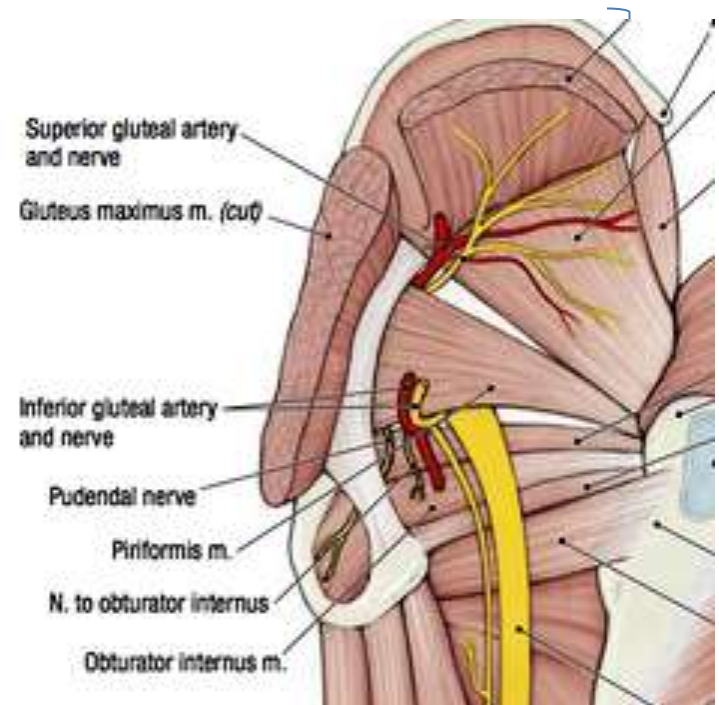
Arteries of gluteal region

1 -Superior gluteal artery

- Branch of *internal iliac artery*.
- It enters gluteal region through the **upper part of greater sciatic foramen** above piriformis.

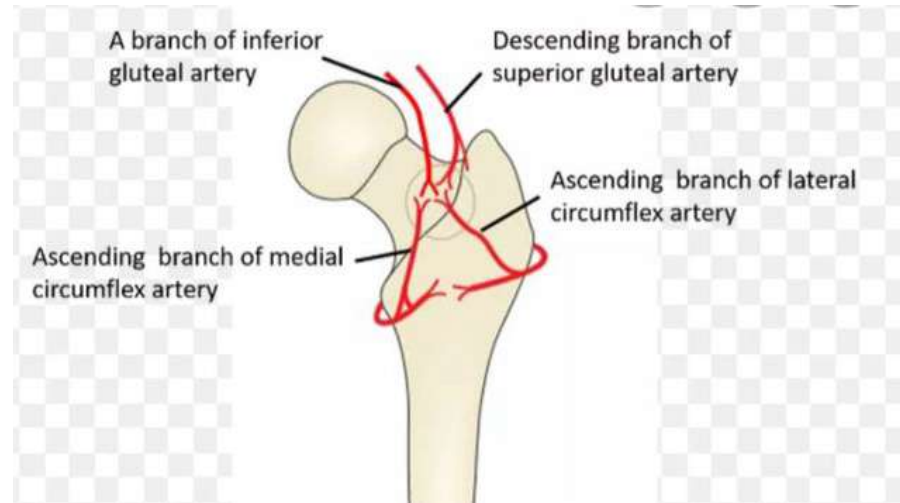
2 -Inferior gluteal artery

- Branch of *internal iliac artery*
- It enters gluteal region through the **lower part of greater sciatic foramen** below piriformis.



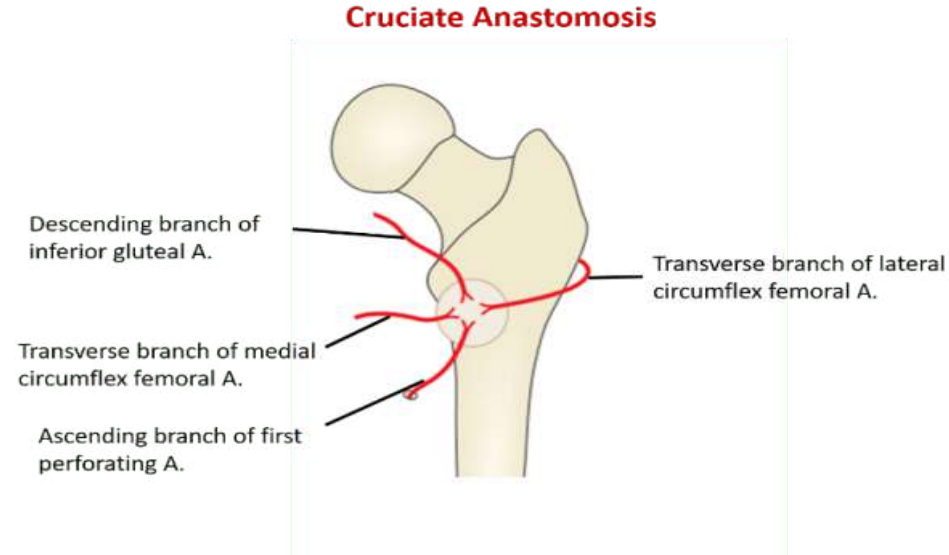
Trochanteric anastomosis

- Provides the **main blood supply to the head of the femur.**
 - The nutrient arteries pass along the femoral neck under the capsule.
- ☐ **The following arteries take part in the anastomosis:**
- 1- **Superior gluteal artery**
 - 2- **Inferior gluteal artery**
 - 3- **Medial femoral circumflex artery**
 - 4- **Lateral femoral circumflex artery**



The Cruciate anastomosis

- Situated at the level of ***lesser trochanter of the femur***.
 - Together with trochanteric anastomosis provides a ***connection between internal iliac and femoral arteries***.
 - ***It supplies blood to LL in case of ligation of femoral artery.***
- ❑ ***The following arteries take part in anastomosis:***
- 1) Inferior gluteal A.
 - 2) Medial femoral circumflex A.
 - 3) Lateral femoral circumflex A.
 - 4) **The first perforating artery**(from profunda femoris).





Formative

All of the following muscles are lateral rotators of the thigh at the hip joint EXCEPT:

- a. Gluteus medius
- b. Gluteus maximus
- c. Superior and inferior gemelli
- d. Quadratus femoris

The nerve innervating the tensor fasciae latae muscle is

- | | |
|---------------------|---------------------|
| A. Superior gluteal | B. Inferior gluteal |
| C. Femoral | D. Obturator |

- **Following muscles are supplied by superior gluteal nerve EXCEPT:**

- a. Gluteus maximus
- b. Gluteus medius
- c. Gluteus minimus
- d. Tensor fascia lata

- **Which of the following muscles is/are hip abductor?**

- a. Gluteus Maximus
- b. Gluteus Medius
- c. Gluteus Medius/Minimus
- d. Gluteus Maximus /Medius.

- Which of the following arteries does not share in arterial cruciate anastomosis?
 - a. Inferior gluteal A.
 - b. Medial femoral circumflex A.
 - c. Lateral femoral circumflex A.
 - d. Superior gluteal



أرخ يدك بالصدقة ترخي حبال المصائب
من على عنقك وأعلم ان حاجتك الى اجر
الصدقة أشد من حاجة ممن تتصدق عليه
"ابن القيم"